

QA DISPOSITION REPORT burn-in parametric screening

Component	LOT213-C0256	Lot	LOT213	Board	LOT213-B04
Socket	r0 c0	Model version	dataset-v1.1 / ModuleA-d13e1ef / ModuleB-2a27c092619:01		

DECISION: FIXTURE SUSPECT

LAYER 1 — deterministic evidence (every value cites its comparison basis)

Absolute limit	PASS (54.90 mA vs 65.0 mA max, supply_current_ma)
Lot-relative level	+4.5 robust sigma vs lot median on leakage_na (median 31.69 nA, sigma 7.40 nA, n=496, MAD estimator; AEC-Q001 IQR/1.35 gives 7.43 nA)
Drift 0h to 24h	+8.174 uA (+48.6%) on iddq_ua; lot median drift +0.584 uA (+4.5%), part is +12.3 robust sigma on the lot's drift distribution
Drift rate	0.3406 uA/h (lot median 0.0243 uA/h, 14.0x)
Max interval jump	+16.5 robust sigma on iddq_ua; largest interval change is 6.0x the median of the others (concentrated -> step)
Predicted 168h	25.013 uA (iddq_ua, from 0h and 24h only)
95% upper bound	32.003 uA -> EXCEEDS lot-derived safe limit 31.321 uA
Joint anomaly	Mahalanobis D2 = 41.2 (good-part 99th pct = 15.3); largest contribution leakage_na at 62% of the total
Leakage/delay sign	leakage +4.5 sigma, delay -0.6 sigma -- consistent with the lot's process locus
Spatial check	52 of 64 components on board LOT213-B04 flagged (expected 1.7, p=3.0e-70) -- CLUSTERED, suspect fixture
Data quality	all checkpoints valid
Primary reason	anomaly clusters by board position, not by component
Secondary	discrete step between checkpoints
Action	Do NOT reject the component on this evidence. Raise a fixture/chamber investigation for the affected board and re-test these positions in a different socket.

LAYER 2 — model attribution

detector	normalised strength	fired	feature (Module B)	SHAP (uA)
L3b_Q	28.77	YES	v24	+8.555
L3b_T2	15.49	YES	d1	+1.256
L4d_AutoEnc	7.54	YES	r1	+0.545
L2_both	4.84	YES	lotmed_t24	+0.149
L3c_kNN	4.03	YES	lotmed_t0	+0.138

Detector strength is distance past that detector's own operating threshold (1.0 = exactly at it). SHAP values are exact TreeSHAP from the LightGBM drift model, in target units. **Caveat:** v0 and v24 are 0.99 correlated, so SHAP credit between them is not identifiable — a Huber model of equal accuracy attributes it the other way round. Treat Layer 2 as supporting evidence; Layer 1 is the auditable record.

COUNTERFACTUAL

No single-parameter change clears the rules; the evidence is distributed across several measurements.

LAYER 3 — failure-mechanism HYPOTHESIS (for the FA engineer; not a diagnosis)

chamber thermal gradient or socket contact degradation [confidence: well-founded]
Because: 52 flagged components share board LOT213-B04, and the leakage/delay shift has the thermal sign (leaky and slow) rather than the process sign
Caveat: Same sign argument, used the other way. HEAT raises leakage and SLOWS the part; the process corner makes leaky parts FAST. So a fixture artifact and a process outlier push the leakage/delay correlation in OPPOSITE directions and are separable by the sign of the shift, before any spatial evidence is considered. The board clustering is then confirmatory rather than the sole basis.
Suggested next test: map the chamber with a calibrated thermocouple board; re-test the affected positions in a different socket

SIGN-OFF

Inspector		Date	
Disposition		Signature	
Notes			

Decision support only. Final rejection remains under approved engineering rules and QA control; this system detects, ranks, explains and forecasts. The model never overrides a hard engineering limit in either direction.